

AI Assignment – Python Variables (Solved)

1. What is a variable?

Answer: A variable is a named memory location used to store data during program execution.

2. How do you declare a variable in Python?

Answer: Python uses assignment. Example: `x = 5`

3. Explain data types.

Answer: Data types define the kind of value a variable can store such as int, float, string, list, tuple, or boolean.

4. What is variable scope?

Answer: Variable scope defines where a variable is accessible. Python has local scope, global scope, and enclosed scope.

5. Example of reassignment.

Answer: `x = 10; x = 20` (Value of x changes from 10 to 20)

6. Naming rules for variables.

Answer: Variable names must not start with a number, cannot contain spaces, and may use letters, digits, and underscores.

7. What happens if you use an undefined variable?

Answer: Python raises a `NameError` because the variable does not exist.

8. What is dynamic typing?

Answer: Dynamic typing means the type of a variable can change at runtime depending on the assigned value.

9. Describe mutability of lists.

Answer: Lists are mutable, meaning their elements can be added, removed, or changed after creation.

10. What is shadowing in nested scopes?

Answer: Shadowing occurs when a variable in an inner scope has the same name as one in an outer scope, hiding the outer variable.